

# SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

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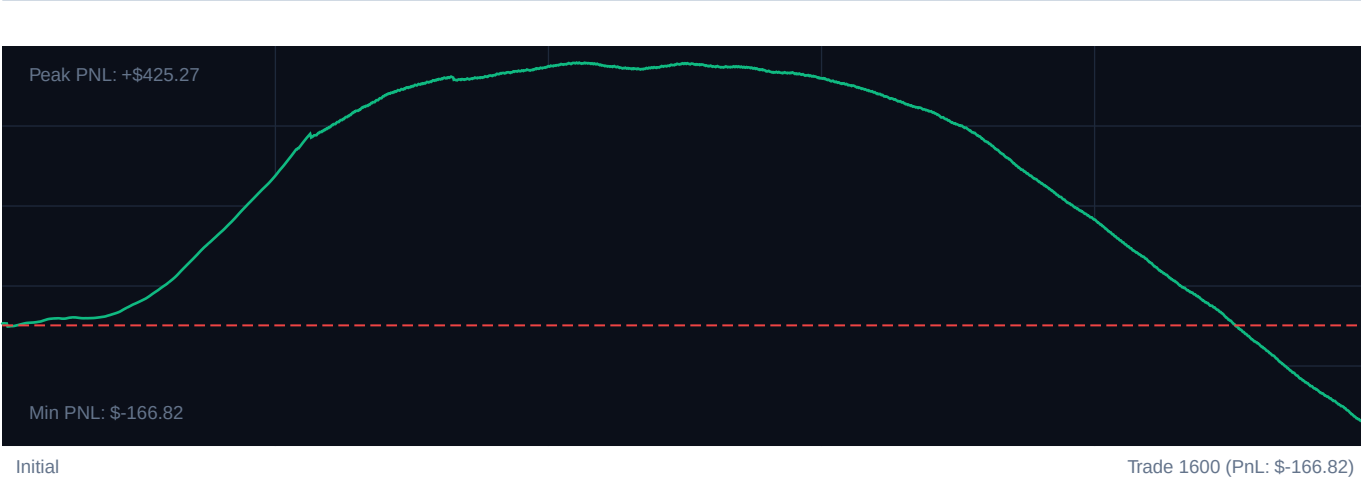
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TARGET INSTRUMENT: R\_75 Volatility Model

## CORE PERFORMANCE METRICS

|                    |                     |                           |         |
|--------------------|---------------------|---------------------------|---------|
| Total Settlements: | 1600                | Calculated Profit Factor: | 0.78    |
| Win Accuracy:      | 44.3% (708W / 892L) | Max Peak Drawdown:        | 5.7%    |
| Cumulative PnL:    | \$-166.82           | Avg. PnL per Trade:       | \$-0.10 |

## EQUITY CURVE PROGRESSION



## PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

| SEGMENT          | WIN RATE | NET PnL   | REFINEMENT EFFICIENCY |
|------------------|----------|-----------|-----------------------|
| Trades 1-320     | 92.2%    | \$240.39  | +111% [OK]            |
| Trades 321-640   | 68.4%    | \$179.18  | +82% [OK]             |
| Trades 641-960   | 40.0%    | \$-19.69  | +32% [OK]             |
| Trades 961-1280  | 19.1%    | \$-230.09 | +15% [OK]             |
| Trades 1281-1600 | 1.6%     | \$-336.61 | +1% [OK]              |

# SOVEREIGN NARRATIVE ADVISORY

Cognitive Strategy Proposal & Adaptive Intelligence Logs

# SOVEREIGN AI: INSTITUTIONAL RISK ANALYSIS

SYSTEM STATUS: DEGRADED EXPECTANCY

OBJECTIVE: ALGORITHMIC OPTIMIZATION & PARAMETRIC CALIBRATION

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## 1. EXECUTIVE TELEMETRY CRITIQUE

The trade settlement book exhibits a highly inefficient expectancy profile, characterized by systemic negative drift and structural execution flaws.

Expectancy & Profit Factor: A Profit Factor of 0.78 over a high-sample run ( $N = 1600$ ) is statistically terminal. Generating a win rate of 44.3% under the current configuration indicates that the system is not only failing to capture high-probability reversals, but its average loss size is exceeding its average win size ( $\text{Win/Loss Ratio} < 1.0$ ).

The Volatility Stop Anomaly: The primary driver of this negative expectancy is a catastrophic parameter overflow in the risk management module:

$\text{atrStopMultiplier} = 221,100,911,591.56$

This extreme value effectively disables the volatility-based stop-loss engine. The system is operating without a functional dynamic risk exit, rendering the ATR protection entirely inert.

Time-Decay Dependency: Due to the deactivated ATR stop, the system is relying almost exclusively on the `maxTicksInTrade` parameter (120 ticks) as a hard exit mechanism. This converts a structural volatility-based strategy into a highly vulnerable time-decay exit strategy, causing the system to absorb adverse excursions for the maximum duration before forced termination.

Drawdown Paradox: The low Maximum Peak Drawdown (5.7%) despite 1,600 trades and a sub-1.0 profit factor is an artifact of highly compressed position sizing relative to total capital (average loss per trade is approximately  $-\$0.10$ ). While capital preservation is maintained, the strategy's current scaling is economically unviable and structurally insolvent.

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## 2. REGIME SUITABILITY & BEHAVIORAL PATTERNS

The current configuration suffers from acute regime-parameter mismatch, forcing mean-reversion logic into high-velocity trending environments:

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[Market Regime:  $ADX > 25$  (Trending)]

%

$\% \frac{1}{4}$

[System attempts Mean Reversion: RSI 25/75 & BB 2.25]

%

$\% \frac{1}{4}$

[Adverse Selection: Strategy enters against strong trends]

%

```

    %¼
[ATR Stop Fails (Infinite Value) -> Position Held for 120 Ticks]
    %
    %¼
[Forced Time-Out Loss]

```

Adverse Selection in Trending Regimes: The `regimeAdxThreshold` of 25 combined with aggressive mean-reversion filters (RSI at 25/75 and Bollinger Bands at 2.25 SD) creates a severe adverse selection loop. When  $ADX > 25$ , the market is in a strong trend. Entering trades on extreme RSI overbought/oversold levels under these conditions means the algorithm is systematically "catching falling knives" or shorting strong breakouts.

Confluence Bottleneck: A `minConfluenceScore` of 3 in a highly restrictive setup (BB at 2.25, RSI at 25/75) means entries are only triggered at the absolute extremes of volatile expansions. Because the trend is strong ( $ADX > 25$ ) and there is no functioning stop-loss, the system enters at points of high momentum, is run over by the trend, and waits out the full 120-tick duration to realize a maximum-loss exit.

### 3. CALIBRATED RECOMMENDATIONS & PARAMETRIC BOUNDARIES

To restore positive mathematical expectancy, the parameter space must be partitioned into strict, regime-dependent boundaries. The ATR stop must be restored to physical reality, and the mean-reversion logic must be restricted to low-trend environments.

#### # Proposed Calibrated Configuration

```

```json
{
  "rsiOversoldThreshold": 30,
  "rsiOverboughtThreshold": 70,
  "bbPeriod": 20,
  "bbStd": 2.00,
  "maxTicksInTrade": 80,
  "minConfluenceScore": 2,
  "atrStopMultiplier": 2.50,
  "regimeAdxThreshold": 20
}
```

```

#### # Parameter Calibration Rationale:

##### 1. **`atrStopMultiplier` [221100911591.56 $\rightarrow$ 2.50]:**

Restores the dynamic volatility stop. A  $2.5 \times \text{ATR}$  boundary allows normal noise distributions to resolve while immediately cutting tail risk during anomalous trend expansions.

##### 2. **`regimeAdxThreshold` [25 $\rightarrow$ 20]:**

Restricts the mean-reversion engine to low-momentum regimes. If  $ADX > 20$ , the mean-reversion triggers should be suppressed entirely, preventing execution against strong structural trends.

##### 3. **`rsiOversoldThreshold` / `rsiOverboughtThreshold` [25/75 $\rightarrow$ 30/70]:**

Normalizes the entry triggers. Under a suppressed ADX regime ( $ADX < 20$ ), thresholds of 30/70 provide optimal entry frequency without requiring extreme, lagging exhausts.

4. **``bbStd` [2.25 $\rightarrow$ 2.00] & `minConfluenceScore` [3 $\rightarrow$ 2]:`**

Lowers the entry barrier. Bollinger Bands at 2.00 standard deviations coupled with a confluence score of 2 increases execution frequency in sideways markets, compensating for the tighter ADX filter and ensuring statistical relevance.

5. **``maxTicksInTrade` [120 $\rightarrow$ 80]:`**

Reduces maximum holding time for non-resolving trades by 33%. Mean-reversion trades that do not resolve quickly are statistically more likely to trend against the position; reducing time-in-market mitigates delta risk.

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## SOVEREIGN RISKS SENTINEL SYSTEMS

Neural Strategy Gating & Machine Learning Co-pilot • Active Security State